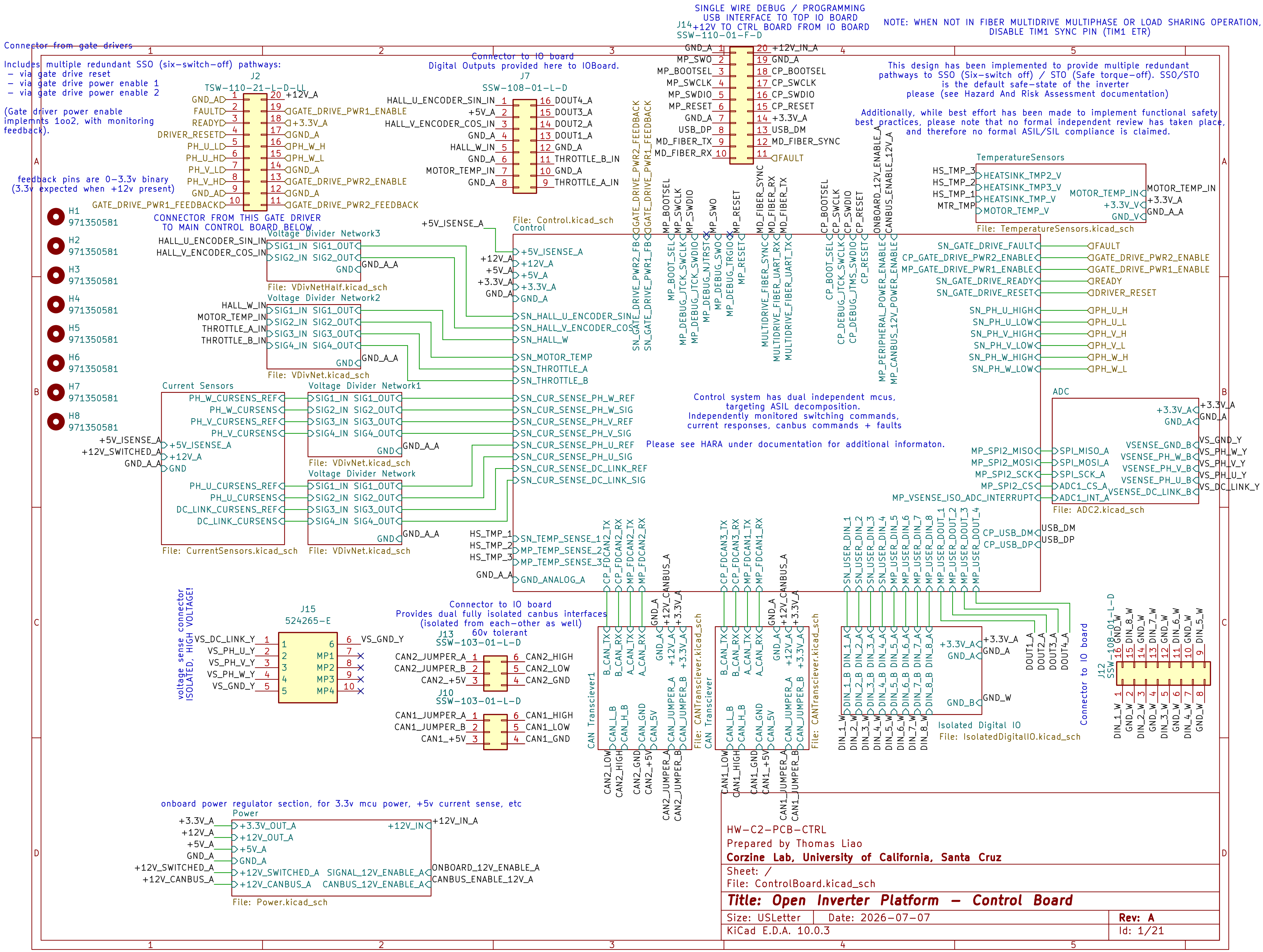
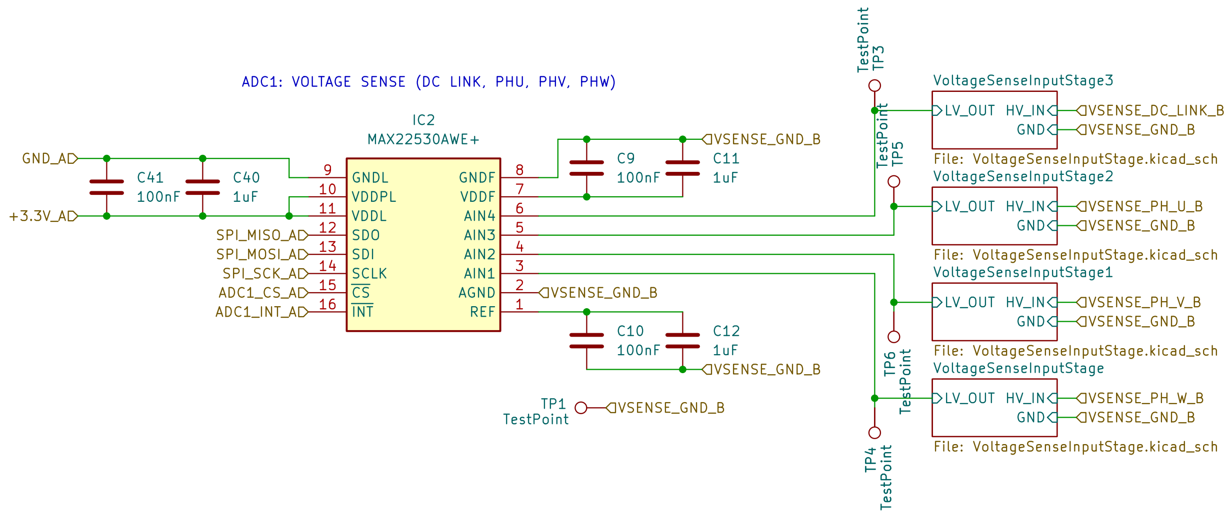

SINGLE WIRE DEBUG / PROGRAMMING
USB INTERFACE TO TOP IO BOARD
+12V TO CTRL BOARD FROM IO BOARD
NOTE: WHEN NOT IN FIBER MULTIDRIVE MULTIPHASE OR LOAD SHARING OPERATION, DISABLE TIM1 SYNC PIN (TIM1 ETR)
Connector from gate drivers
Includes multiple redundant SSO (six-switch-off) pathways:
- via gate drive reset
- via gate drive power enable 1
- via gate drive power enable 2
(Gate driver power enable implemnts 1oo2, with monitoring feedback).
feedback pins are 0-3.3v binary (3.3v expected when +12v present)
CONNECTOR FROM THIS GATE DRIVER TO MAIN CONTROL BOARD BELOW
Connector to IO board
Digital Outputs provided here to IOBoard.
This design has been implemented to provide multiple redundant pathways to SSO (Six-switch off) / STO (Safe torque-off). SSO/STO is the default safe-state of the inverter please (see Hazard And Risk Assessment documentation)
Additionally, while best effort has been made to implement functional safety best practices, please note that no formal independent review has taken place, and therefore no formal ASIL/SIL compliance is claimed.
Control system has dual independent mcus, targeting ASIL decomposition. Independently monitored switching commands, current responses, canbus commands + faults
Please see HARA under documentation for additional informaton.
voltage sense connector ISOLATED, HIGH VOLTAGE!
Connector to IO board
Provides dual fully isolated canbus interfaces (isolated from each-other as well) 60v tolerant
Connector to IO board
onboard power regulator section, for 3.3v mcu power, +5v current sense, etc
File: Control.kicad_sch
File: TemperatureSensors.kicad_sch
File: ADC2.kicad_sch
File: VDivNetHalf.kicad_sch
File: VDivNet.kicad_sch
File: CurrentSensors.kicad_sch
File: CANTransciever.kicad_sch
File: IsolatedDigitalIO.kicad_sch
File: Power.kicad_sch
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: ControlBoard.kicad_sch
Title: Open Inverter Platform - Control Board
Size: USLetter
Date: 2026-07-07
Rev: A
KiCad E.D.A. 10.0.3
Id: 1/21

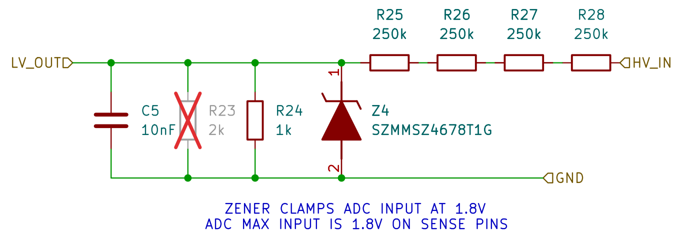


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/
File: ADC2.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 4/21

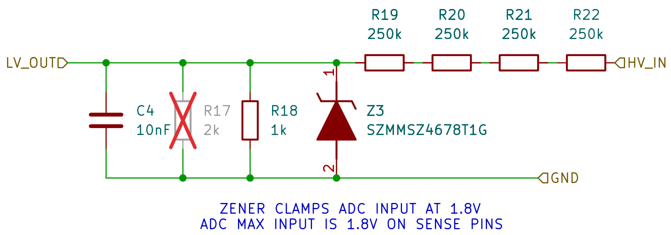


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage/
File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 2/21

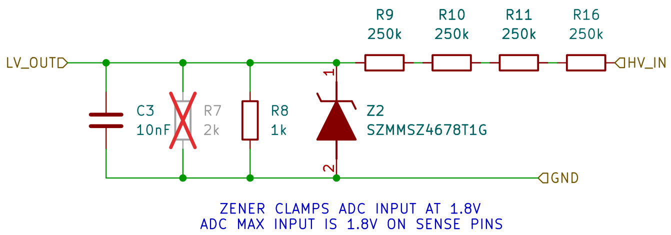


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage1/
File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 3/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

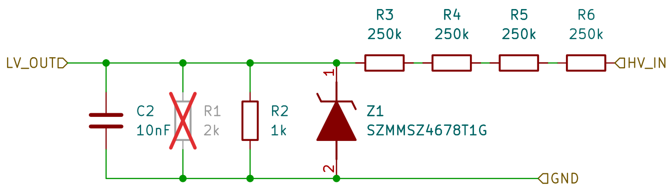
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage2/

File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 5/21

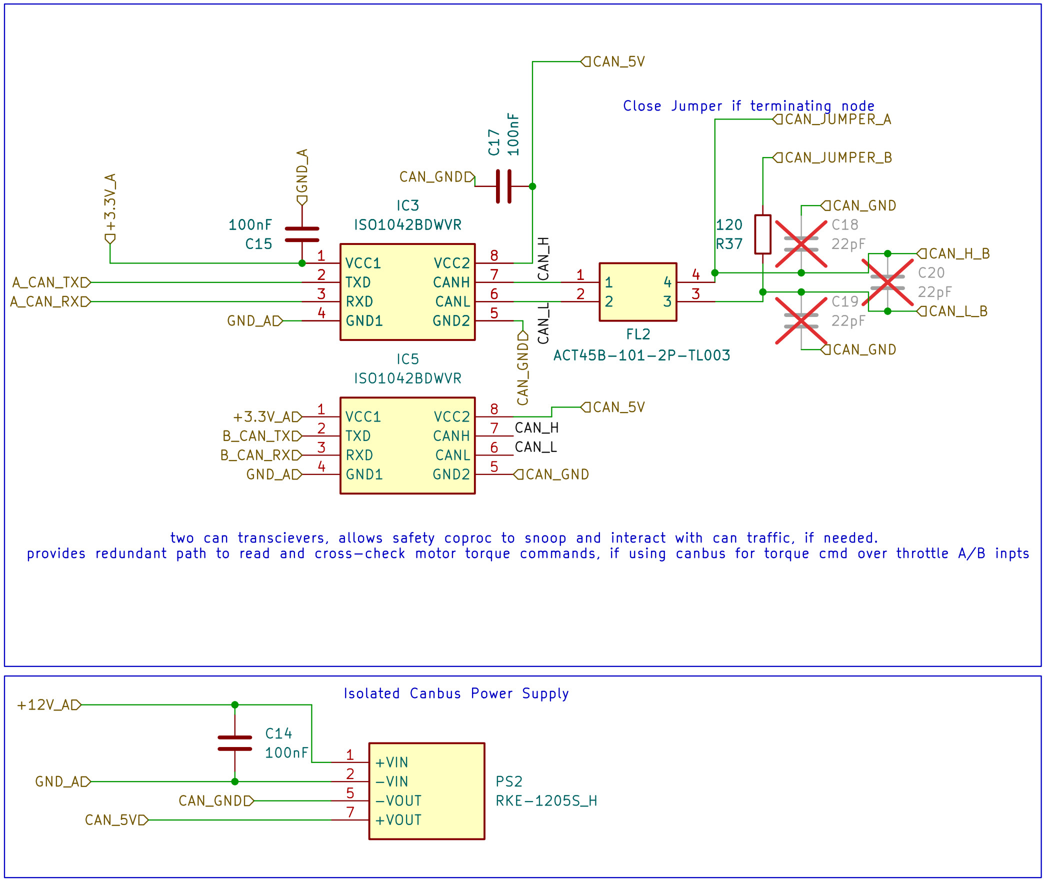


ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage3/
File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform - Control Board

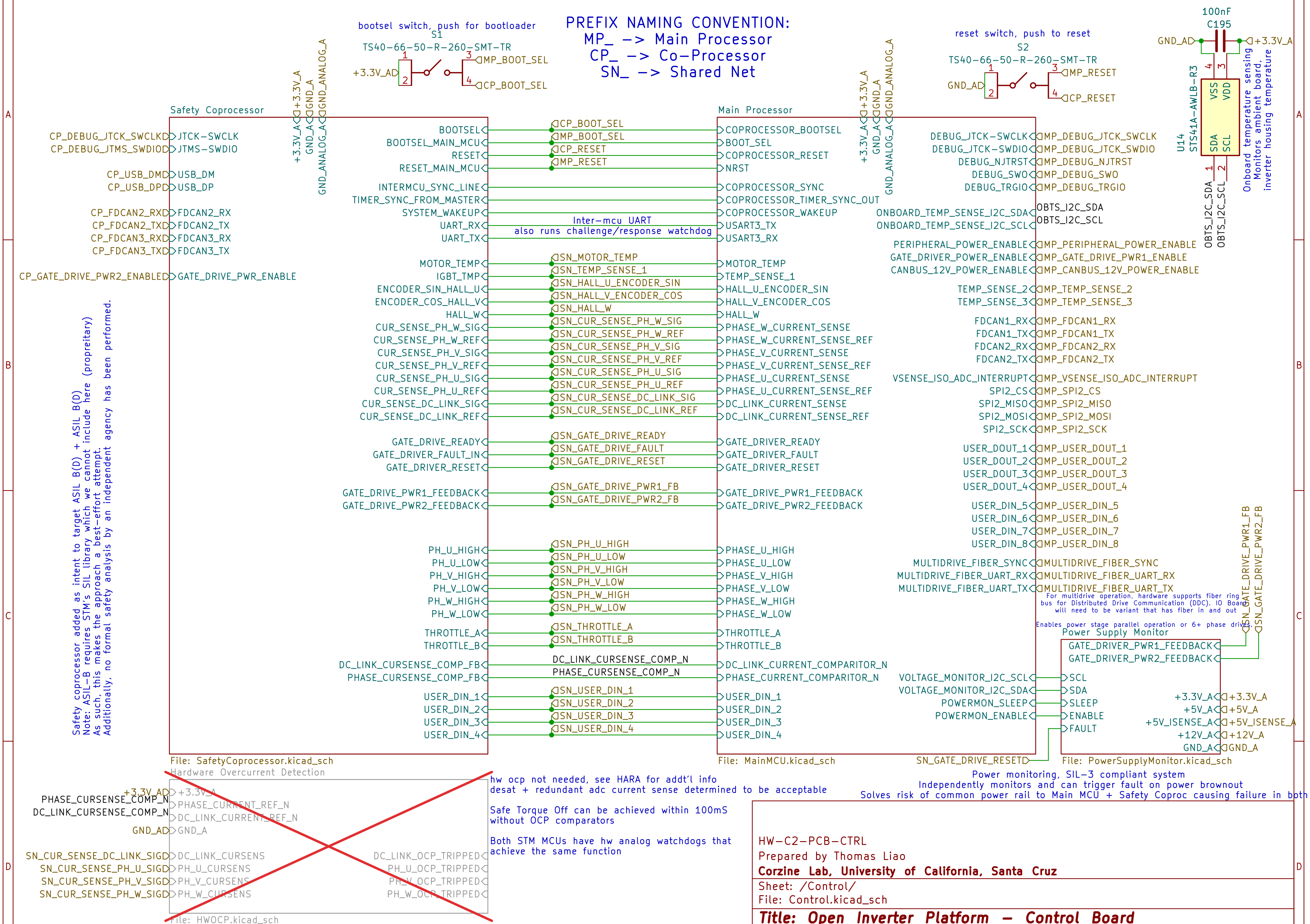
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 6/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /CAN Transciever1/
File: CANTransciever.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 7/21

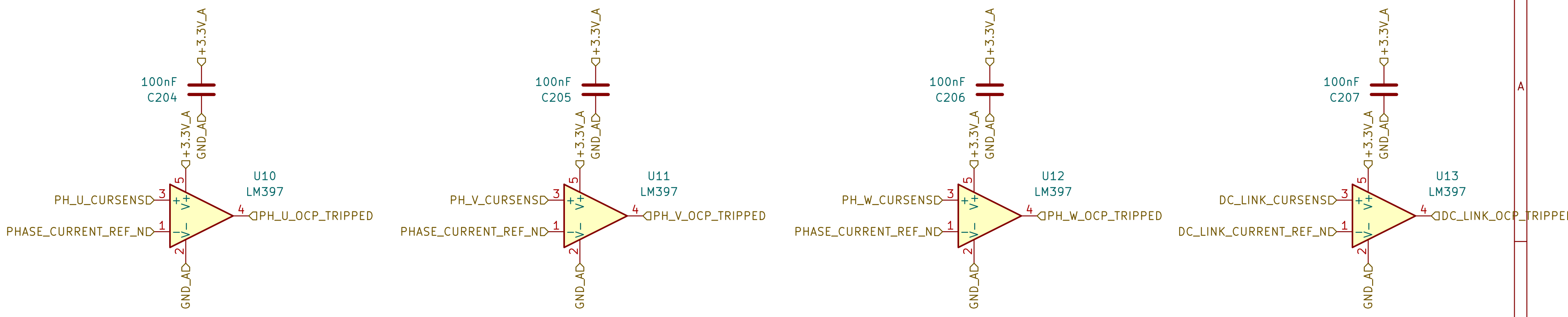


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Control/
File: Control.kicad_sch

Title: Open Inverter Platform - Control Board

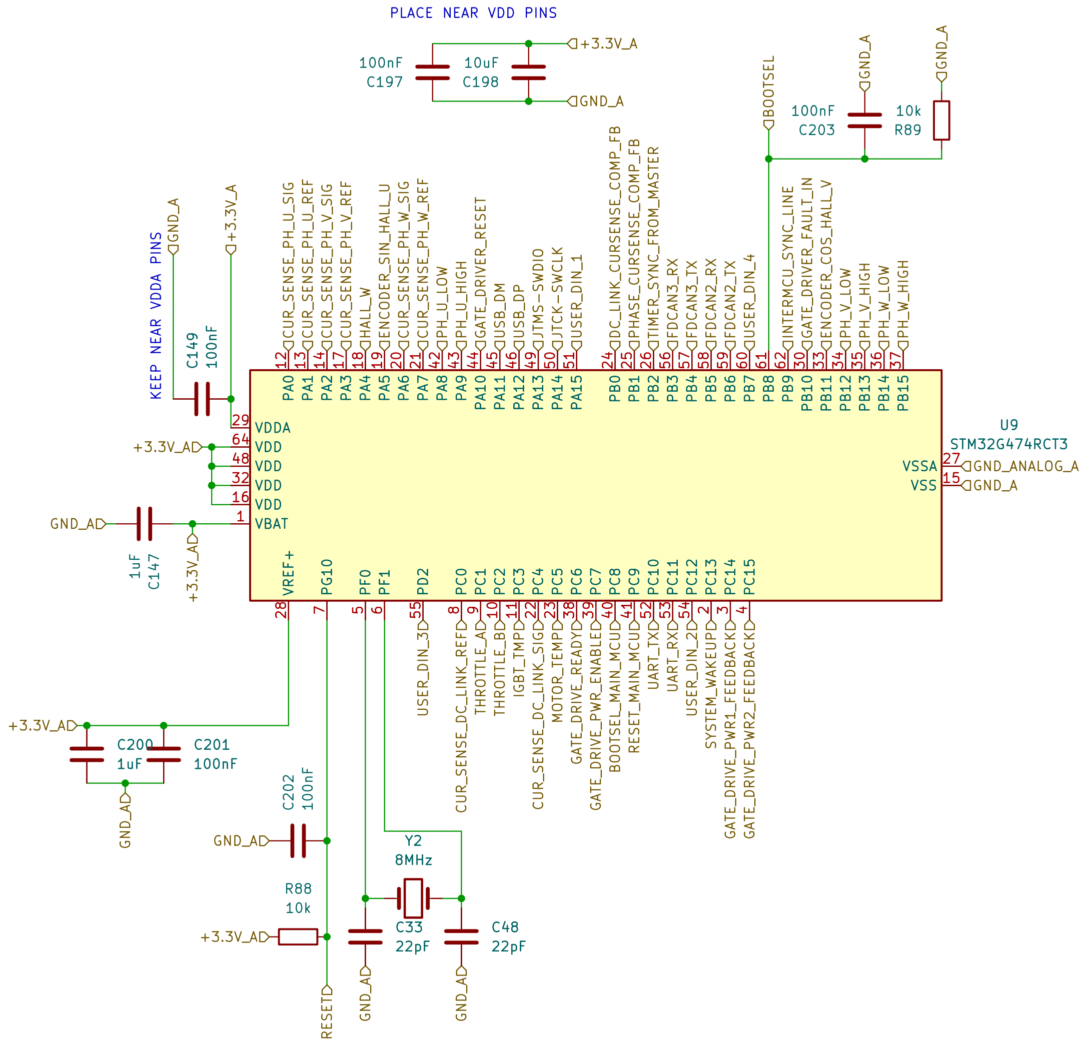
Size: USLetter	Date: 2026-07-07	Rev: A
KiCad E.D.A. 10.0.3		Id: 8/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Hardware Overcurrent Detection/
File: HWOCP.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 10/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

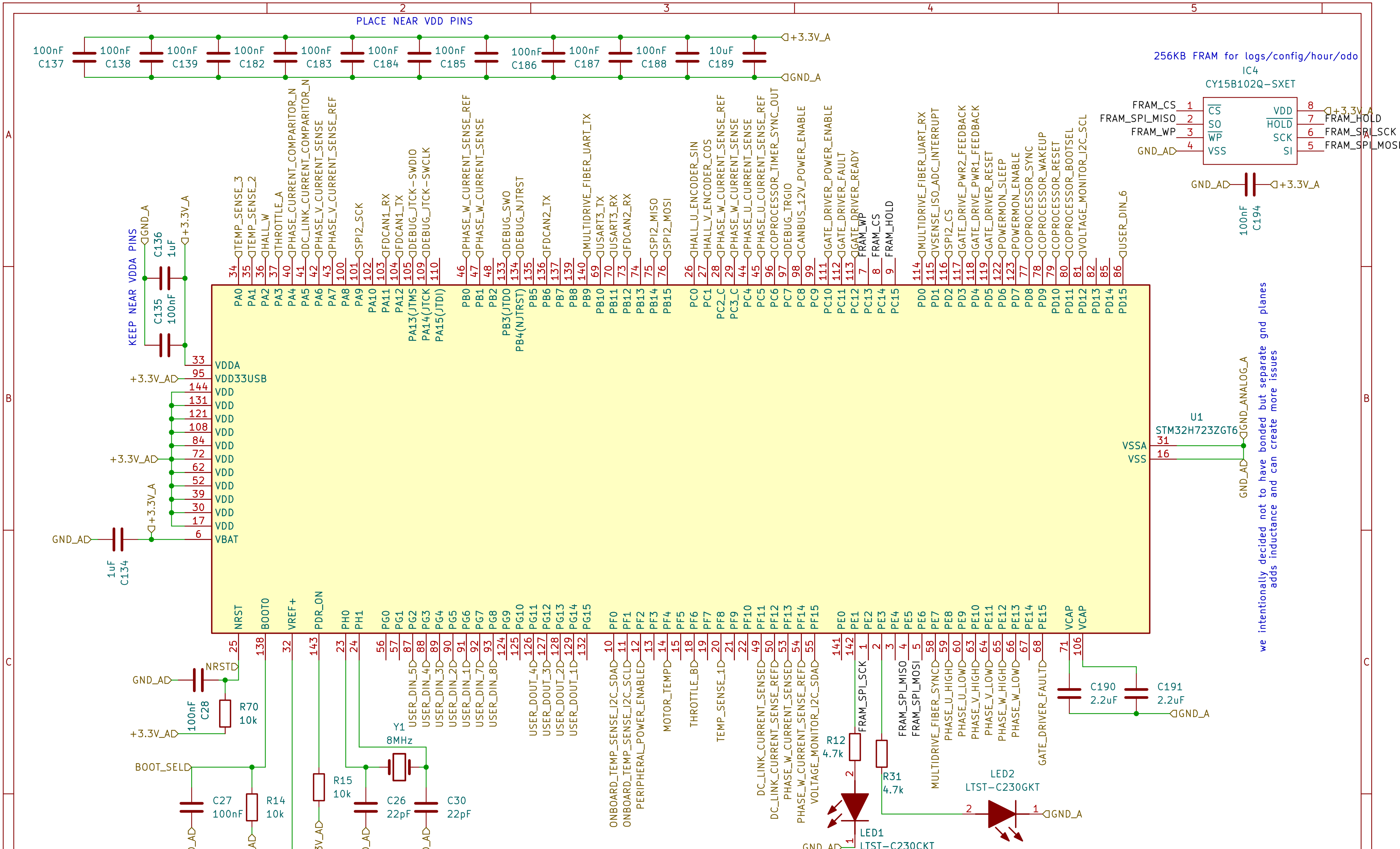
Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Safety Coprocessor/

File: SafetyCoprocessor.kicad_sch

Title: Open Inverter Platform - Control Board

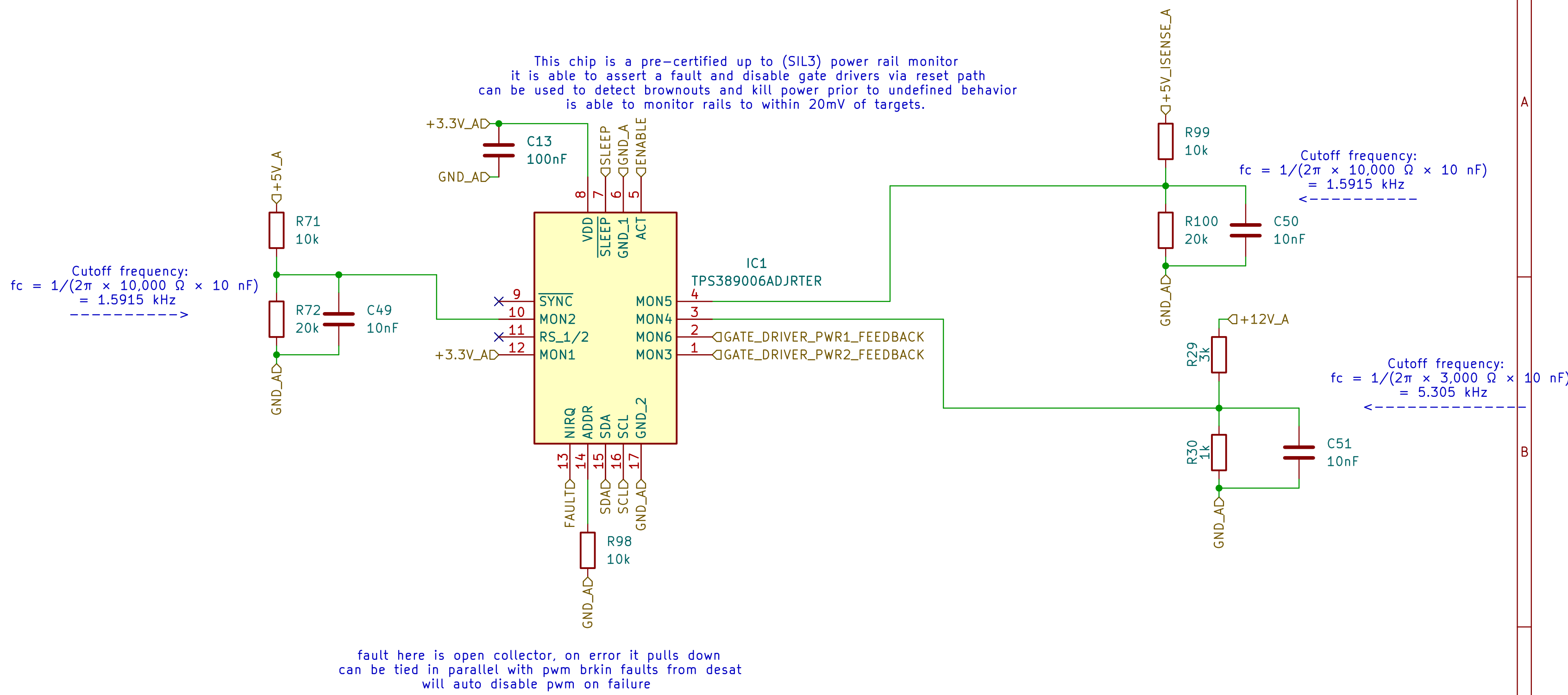
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 16/21



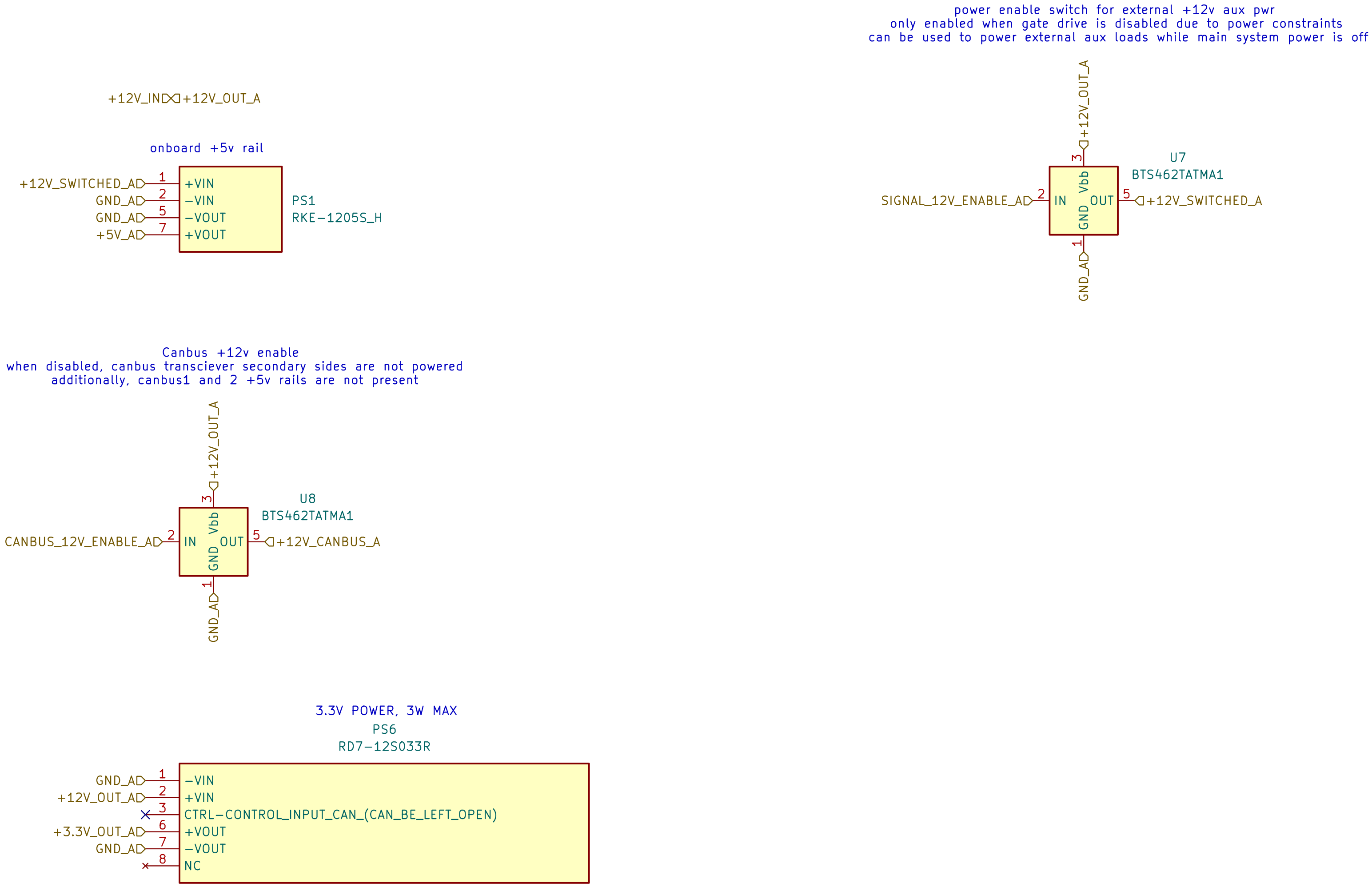
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Main Processor/
File: MainMCU.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 19/21



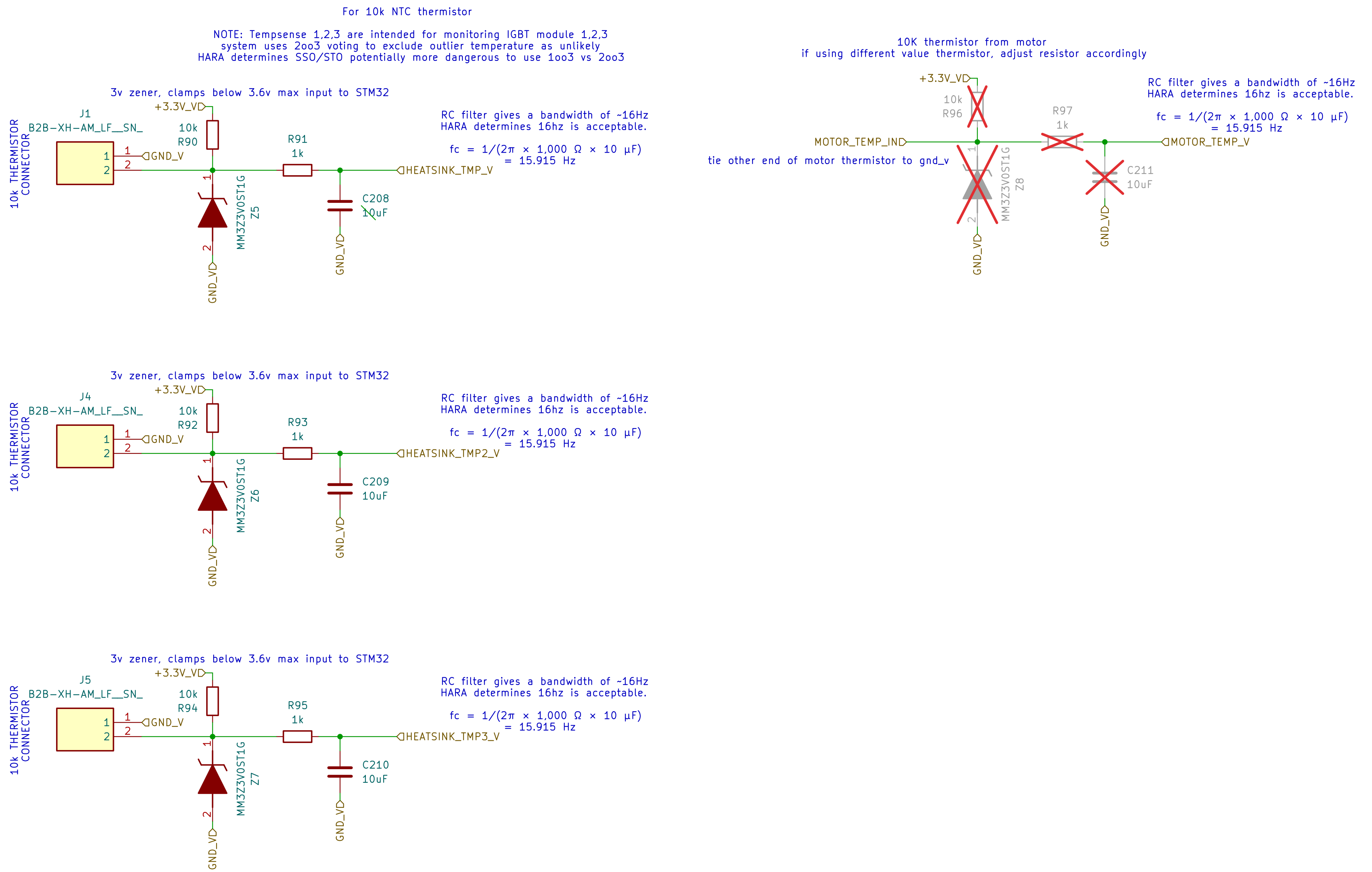
HW−C2−PCB−CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Power Supply Monitor/
File: PowerSupplyMonitor.kicad_sch
Title: Open Inverter Platform − Control Board
Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 20/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Power/
File: Power.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 9/21

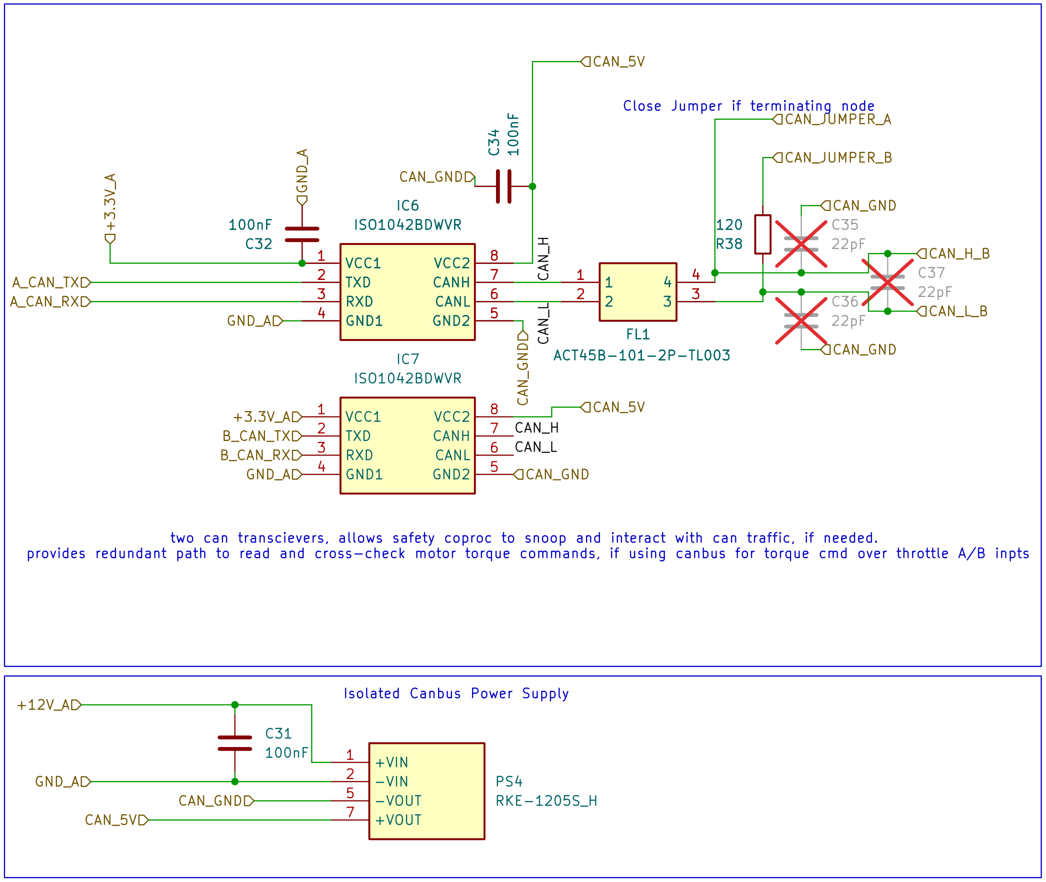


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /TemperatureSensors/
File: TemperatureSensors.kicad_sch

Title: Open Inverter Platform – Control Board

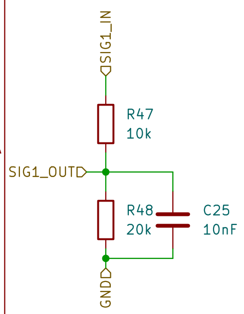
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 9/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /CAN Transciever/
File: CANTransciever.kicad_sch

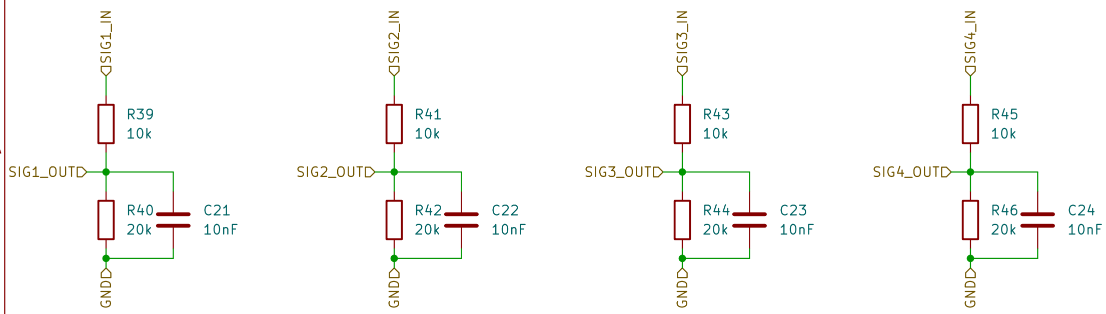
Title: Open Inverter Platform - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 11/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Voltage Divider Network1/
File: VDivNet.kicad_sch
Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 12/21

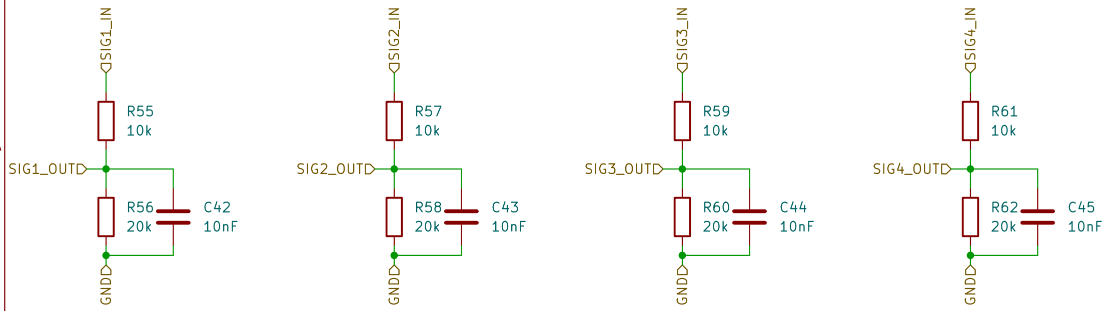


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network/
File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 13/21

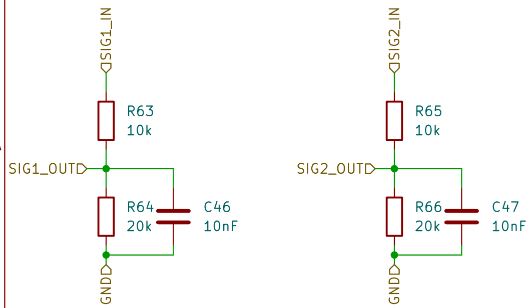


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network2/
File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

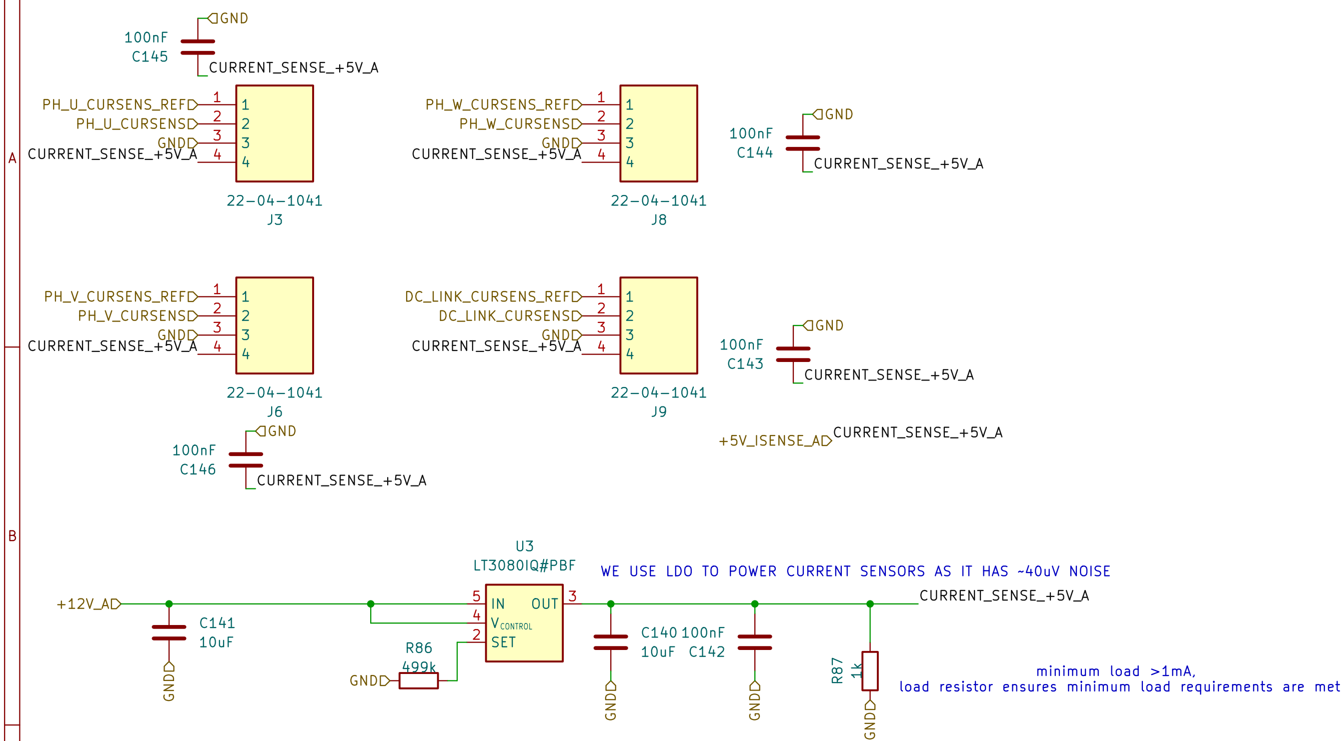
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 14/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Voltage Divider Network3/
File: VDivNetHalf.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 15/21

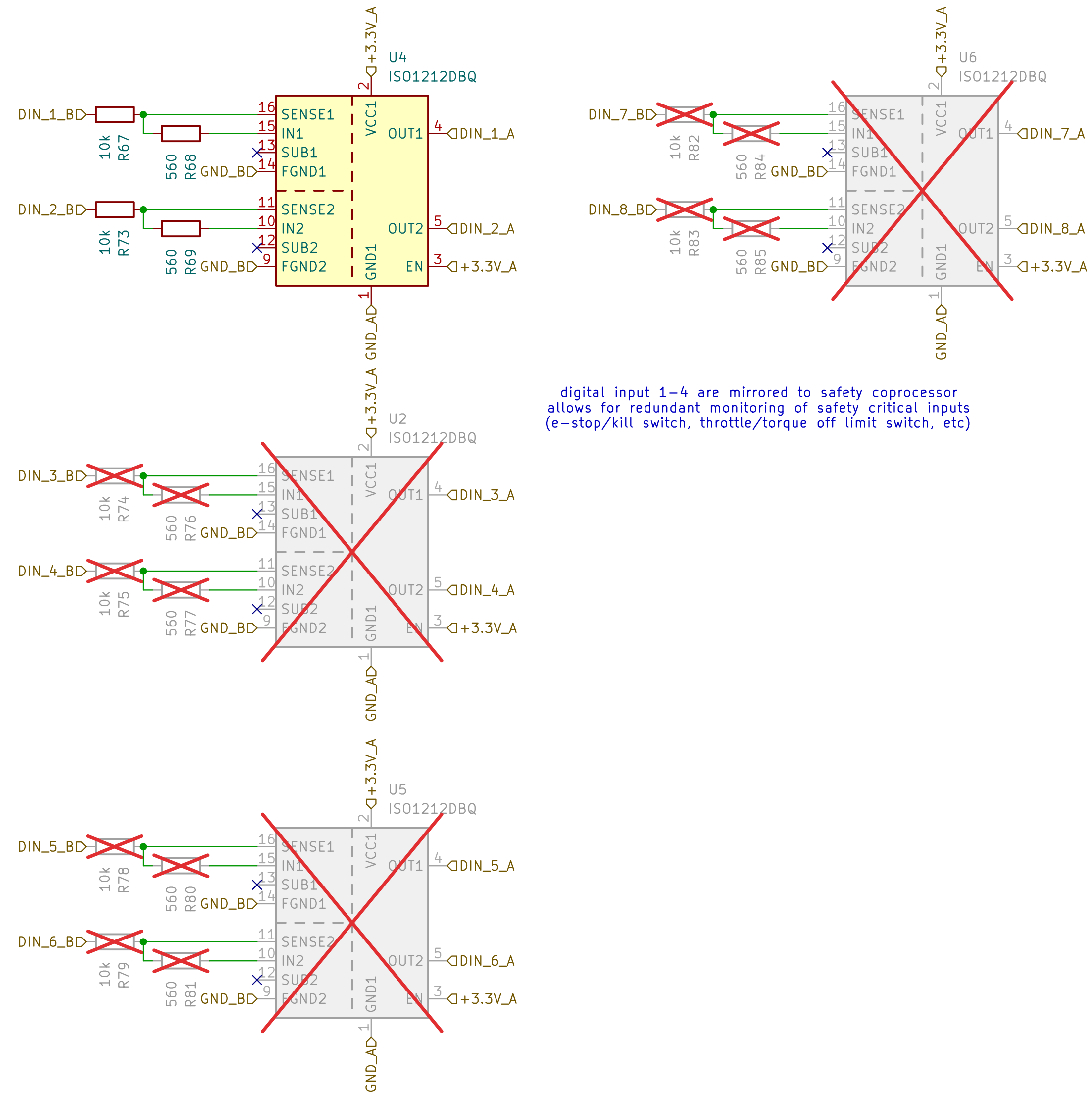


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Current Sensors/
File: CurrentSensors.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 17/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Isolated Digital IO/

File: IsolatedDigitalIO.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 18/21